ACTIVITY

Media Content Integration in Web Applications

Course	IT AppDev 2 – Web Applications
Activity	Website Development Project – Media Content Integration
Coverage	Web media types, image/audio/video integration, responsive and adaptive media, file upload and media management, accessibility and validation
Aligned CLO	CLO1
Recommended Duration	Weeks 2–3
Expected Output	Functional, responsive, accessible, and validated website with integrated media content

Activity Alignment

This activity supports CLO1 by requiring students to deploy and serve media content within a functional web application. The syllabus identifies Weeks 2–3 as the coverage for media content integration and expects a functional web page/application with integrated, responsive, accessible, and validated media content.

I. Activity Overview

Develop a functional, responsive, and accessible website that demonstrates your ability to integrate different types of media content into a web application.

Your website must demonstrate proper use of images, audio, video, responsive media techniques, file uploading, media management, accessibility practices, and HTML validation.

The website should not simply display media files. It must demonstrate that you understand when and how different media formats should be used in a web application.

II. Website Theme

Choose ONE website concept from the following:

- Travel and Tourism Website
- Food and Recipe Website
- School/Student Organization Website
- Local Culture and Heritage Website
- Photography Portfolio
- Music and Entertainment Website
- Sports Website
- Community Information Website
- Product Showcase Website
- Personal Portfolio Website

III. Required Website Components

A. Home Page

Create a visually organized home page containing:

- Website title/logo
- Navigation menu
- Introduction or welcome section

- At least one optimized image
- At least one responsive media element
- Appropriate headings and text
- Footer

The page should clearly communicate the purpose of the website.

B. Image Integration

Your website must include at least three images.

For each image:

- Use an appropriate image format such as JPEG, PNG, WebP, or SVG.
- Provide meaningful alt text.
- Use appropriate dimensions.
- Avoid unnecessarily large image files.
- Apply CSS for proper presentation.
- Ensure images remain usable on mobile devices.

At least one image must demonstrate a responsive technique such as:

- Responsive width
- srcset
- <picture>
- CSS-based responsive sizing

Example:

```
<picture>
    <source media="(max-width: 600px)" srcset="images/mobile.webp">
    <source media="(max-width: 1200px)" srcset="images/tablet.webp">
    <img src="images/desktop.webp"
         alt="Description of the featured image">
</picture>
```

C. Audio Integration

Integrate at least one audio file into your website.

The audio may be:

- Background narration
- Interview
- Music
- Podcast excerpt
- Environmental sound
- Voice introduction

Use the HTML5 <audio> element.

```
<audio controls>
    <source src="media/audio.mp3" type="audio/mpeg">
    Your browser does not support the audio element.
</audio>
```

Requirement

Do not use audio merely as decoration. Provide a meaningful purpose for the audio content.

D. Video Integration

Integrate at least one video.

The video may be:

- Website introduction
- Tutorial
- Demonstration
- Travel presentation
- Product demonstration
- Documentary clip
- Promotional video

Use the HTML5 <video> element or an appropriately embedded video service.

```
<video controls width="100%">
    <source src="media/video.mp4" type="video/mp4">
    Your browser does not support the video element.
</video>
```

The video must include appropriate accessibility features where applicable, such as captions or a transcript.

IV. Responsive and Adaptive Media

Your website must work properly on:

- Desktop/laptop
- Tablet
- Smartphone

Use responsive web design techniques.

At minimum, implement:

- Flexible image sizing
- Responsive video
- Responsive audio controls
- CSS media queries
- Flexible containers
- Mobile-friendly navigation

Example:

```
img,
video {
    max-width: 100%;
    height: auto;
}

.media-container {
    width: 100%;
}
```

You must test the website using the browser's responsive/device simulation tools.

Document your testing using screenshots showing the website at different viewport sizes.

V. File Upload Functionality

Create a Media Upload Page that allows a user to upload a media file.

The page should contain:

- File selection field

- Upload button
- Appropriate form labels
- File type restrictions
- File size restrictions
- Upload status/message

Example:

```
<form action="upload.php" method="POST" enctype="multipart/form-data">

    <label for="mediaFile">Select Media File:</label>

    <input type="file"
           id="mediaFile"
           name="mediaFile"
           accept="image/*,audio/*,video/*">

    <button type="submit">Upload Media</button>

</form>
```

If PHP is being used, the server-side upload process should validate the uploaded file before storing it.

VI. Media Management

Create a simple Media Gallery or Media Management Page.

The page should allow users to:

- View uploaded media
- Identify the media type
- Display media appropriately
- Delete media when applicable
- Organize media files logically

Recommended folder structure:

```
project/
│
├── index.php
├── upload.php
├── gallery.php
├── css/
│   └── style.css
│
├── images/
│
├── audio/
│
├── videos/
│
└── uploads/
```

Use meaningful filenames rather than names such as:

- IMG001.jpg
- video123.mp4
- file1.png

Prefer names such as:

- isabela-tourism.webp
- local-food.jpg
- heritage-interview.mp3
- museum-tour.mp4

VII. Media Accessibility

Your website must follow basic web accessibility practices.

Students must demonstrate the following:

Images

Use meaningful alternative text.

```
<img src="images/museum.jpg"
     alt="Historical museum building in Ilagan City">
```

Do not use:

```
<img src="images/museum.jpg" alt="image">
```

Audio

Provide controls and, where appropriate, a text transcript.

Video

Provide:

- Controls
- Captions/subtitles where applicable
- Transcript or alternative description when appropriate

Forms

Every file upload field must have an associated label.

```
<label for="mediaFile">Select media file:</label>
<input type="file" id="mediaFile" name="mediaFile">
```

General Accessibility

Ensure that:

- Text has sufficient contrast.
- Content is readable.
- Navigation is understandable.
- Interactive elements can be accessed using a keyboard.
- Images contain meaningful alternative text.
- Media does not automatically play with sound.

VIII. Media Validation

Before submitting the website, validate your HTML and CSS.

Check for:

- Missing closing tags
- Invalid HTML elements
- Missing alt attributes

- Incorrect media paths
- Invalid attributes
- Broken links
- CSS errors
- Missing form labels

Students should also test the website using browser developer tools.

Functional Testing

Verify that:

- Images load correctly.
- Audio plays correctly.
- Video plays correctly.
- Uploaded files are stored correctly.
- Uploaded files can be displayed.
- Invalid files are rejected.
- Oversized files are rejected.
- Responsive layouts work.
- Media does not overflow the screen.
- Navigation works on mobile devices.

IX. Media Format Documentation

Create a small table on your website or in your documentation identifying the media formats you used.

Media Type	Format Used	Purpose	Reason for Selection
Image	WebP/JPEG/PNG	Website images	Appropriate for web delivery
Audio	MP3/OGG	Narration/music	Compatible with web browsers
Video	MP4/WebM	Video presentation	Suitable for web playback

Students must be able to explain why a particular format was selected.

X. Required Pages

Your completed website should contain at least the following pages:

1. 1. Home
2. 2. About / Information
3. 3. Media Gallery
4. 4. Upload Media
5. 5. Contact / Credits

Additional pages may be created depending on your selected website concept.

XI. Minimum Technical Requirements

Your project must demonstrate:

☐ HTML5

☐ CSS3

☐ Responsive Web Design

☐ HTML5 <img>

☐ HTML5 <audio>

☐ HTML5 <video>

☐ Responsive media techniques

☐ File upload form
☐ Media file validation
☐ Media gallery/management
☐ Accessibility features
☐ Form labels
☐ Alternative text
☐ Proper media organization
☐ HTML/CSS validation
☐ Browser testing

XII. Suggested Development Workflow

Follow this development sequence:

Step 1 – Plan

- Website topic
- Target users
- Required pages
- Types of media to use
- Media formats
- File organization

Step 2 – Design

- Create a simple wireframe showing:
- Navigation
- Content areas
- Media sections
- Gallery
- Upload interface

Step 3 – Develop

- Create the HTML and CSS structure.

Step 4 – Integrate Media

- Images
- Audio
- Video
- Responsive media

Step 5 – Implement Upload

- Create the upload form and implement file validation.

Step 6 – Implement Media Management

- Display uploaded files and organize them according to their media type.

Step 7 – Apply Accessibility

- alt attributes
- Labels
- Captions
- Transcripts

- Keyboard accessibility
- Readability

Step 8 – Test

- Test the application on different screen sizes and browsers.

Step 9 – Validate

- Check HTML, CSS, links, forms, media files, and upload functionality.

Step 10 – Document

- Prepare screenshots and technical documentation explaining your implementation.

XIII. Required Documentation

Submit a short technical documentation containing:

1. Project Title

- Example: “Discover Isabela: A Responsive Tourism and Heritage Website”

2. Project Description

- Explain:
- Purpose of the website
- Target users
- Main features

3. Media Inventory

- List all images, audio, and video used.

4. Media Format Justification

- Explain why each media format was selected.

5. Responsive Design

- Explain how your website adapts to:
- Desktop
- Tablet
- Mobile

6. Accessibility Implementation

- Explain the accessibility features implemented.

7. File Upload and Management

- Explain how files are uploaded, validated, stored, and displayed.

8. Testing Results

- Include screenshots demonstrating:
- Desktop view
- Tablet view
- Mobile view
- Media playback
- Upload functionality
- Validation results

9. Problems Encountered and Solutions

- Identify at least three technical problems and explain how you solved them.

10. Conclusion

- Summarize what you learned from the activity.

XIV. Submission Requirements

Submit the following:

A. Source Code

```
website-project/
├── HTML/PHP files
├── CSS files
├── JavaScript files (if used)
├── images/
├── audio/
├── videos/
└── uploads/
```

B. Technical Documentation

PDF or DOCX containing the required documentation and screenshots.

C. Project Demonstration

Demonstrate the following during the practical presentation:

- Website navigation
- Image integration
- Audio playback
- Video playback
- Responsive behavior
- File upload
- Media management
- Accessibility implementation
- Validation/testing
- Explanation of media format choices

XV. Assessment Criteria

The activity is aligned with the syllabus assessment emphasis on media integration, functionality, optimization, storage, accessibility, deployment, and secure content delivery. The syllabus identifies a 3/4 Proficient level as the rubric standard for CLO1 and targets at least 75% of students meeting the threshold.

Criteria	Weight
Website functionality and navigation	15%
Image integration and optimization	15%
Audio and video integration	15%
Responsive/adaptive media	15%
File upload and media management	15%
Accessibility implementation	10%
Validation and testing	10%
Code organization and documentation	5%
Total	**100%**

Performance Expectation

A proficient submission should demonstrate that the student can integrate media into a functional web application rather than simply embed media files. The completed website should be responsive, accessible, validated, and properly organized.

Course AI Policy Reminder

The course syllabus states that the acceptable AI assistance threshold for student assessment submissions shall not exceed 25%. Students are responsible for ensuring their work complies with the approved classroom AI policy.